Do Now · retrieval

Lesson 095 · CPU Pipelining: Overlap and Hazards

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is pipeline throughput?

Q2

What is a hazard?

Q3

Which CPU feature may reduce RAM accesses?

Q4

A D flip-flop has D=0 and Q=1 just before the active rising edge. What is Q after?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Pipelining splits instruction processing into stages; several instructions can occupy different stages during the same clock cycle.

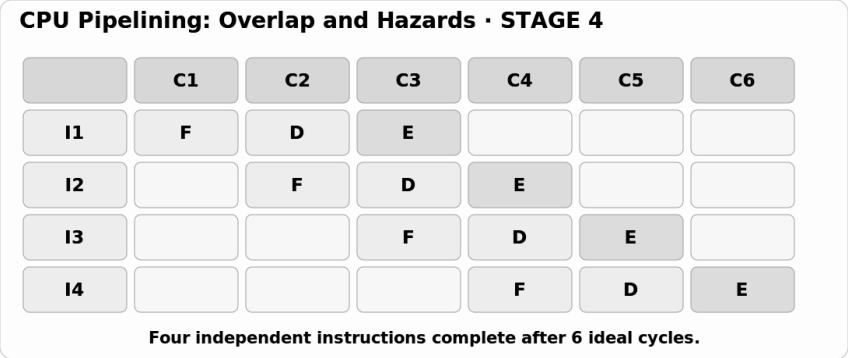
Core idea 2

Explain — With four independent instructions and three one-cycle stages, ideal completion is 3+4−1=6 cycles; serial non-overlap takes 4×3=12.

Core idea 3

Evaluate — Data dependencies, branching and resource conflicts can stall or flush a pipeline. Throughput benefits need ideal assumptions, not automatic latency reductions.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

pipeline — Stages overlapping different instructions.

throughput — Completed work per time.

data hazard — Dependency on unavailable result.

control hazard — Branch uncertainty interrupting fetch.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

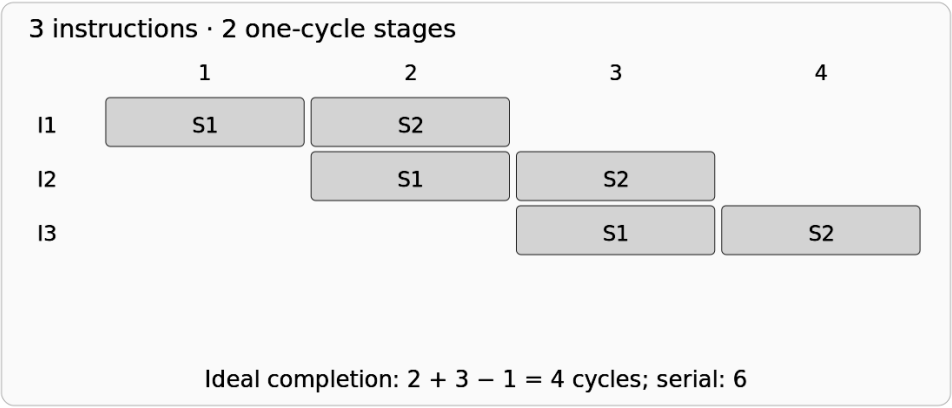
Problem

Three instructions pass through a two-stage pipeline. Compute ideal completion cycles.

Follow the worked method

1. Explain the mechanism — 2+3−1=4 cycles; serial baseline 2×3=6 cycles. — The second task starts stage one while first advances to stage two.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the mechanism
$4+5-1=8$ cycles, compared with $4\times5=20$ serial stage cycles.
Assumes equal one-cycle stages, no resource conflicts or stalls.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Show cycles for I1-I3 in a 3-stage F-D-E pipeline, and finish cycle of I3. [6]

Question 2

Compute ideal pipeline versus serial completion for 6 instructions and 3 stages. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why branch and data hazards can reduce actual improvement, distinguishing latency and throughput. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A pipeline makes each stage finish in less than one cycle by default.

Use key vocabulary from memory

1. Explain pipeline in your own words.

2. Explain throughput in your own words.

3. Explain data hazard in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.